AIRBNB RENTAL ARBITRAGE

Identifying High-Value Rental Opportunities in New York City

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Course	MGMT 59000 AI for Business Analytics - DY2
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Semester	Summer 2026
Submission date	July 26, 2026
Dataset	Airbnb_db.csv | 4,892 NYC listings | 2025

Executive Summary

Business objective. Help Joan identify apartments likely to command more than $120 per night before signing a lease.

Listings	Average price	Above $120	Top borough
4,892	**$138.51**	**66.3%**	**Manhattan**

Key recommendation. Prioritize Manhattan, Brooklyn, and Queens during initial screening. Use review activity, minimum-night policy, and annual availability as secondary factors, then validate the predicted price against rent, occupancy, fees, regulations, and landlord permission.

1. Problem - Data - Insights - Deployment

Problem

Joan needs a fast, repeatable way to understand which listing characteristics are associated with higher nightly prices and to estimate the likely nightly price of a specific opportunity.

Data

The analysis uses 4,892 New York City Airbnb listings. Predictors include borough, minimum nights, total reviews, reviews per month, and annual availability. Price is the target. The id field is retained only as a row identifier and excluded from modeling.

Insights

A Multiple Linear Regression model quantifies the joint relationship between the predictors and nightly price. The app also flags whether a predicted listing clears the $120 threshold.

Deployment

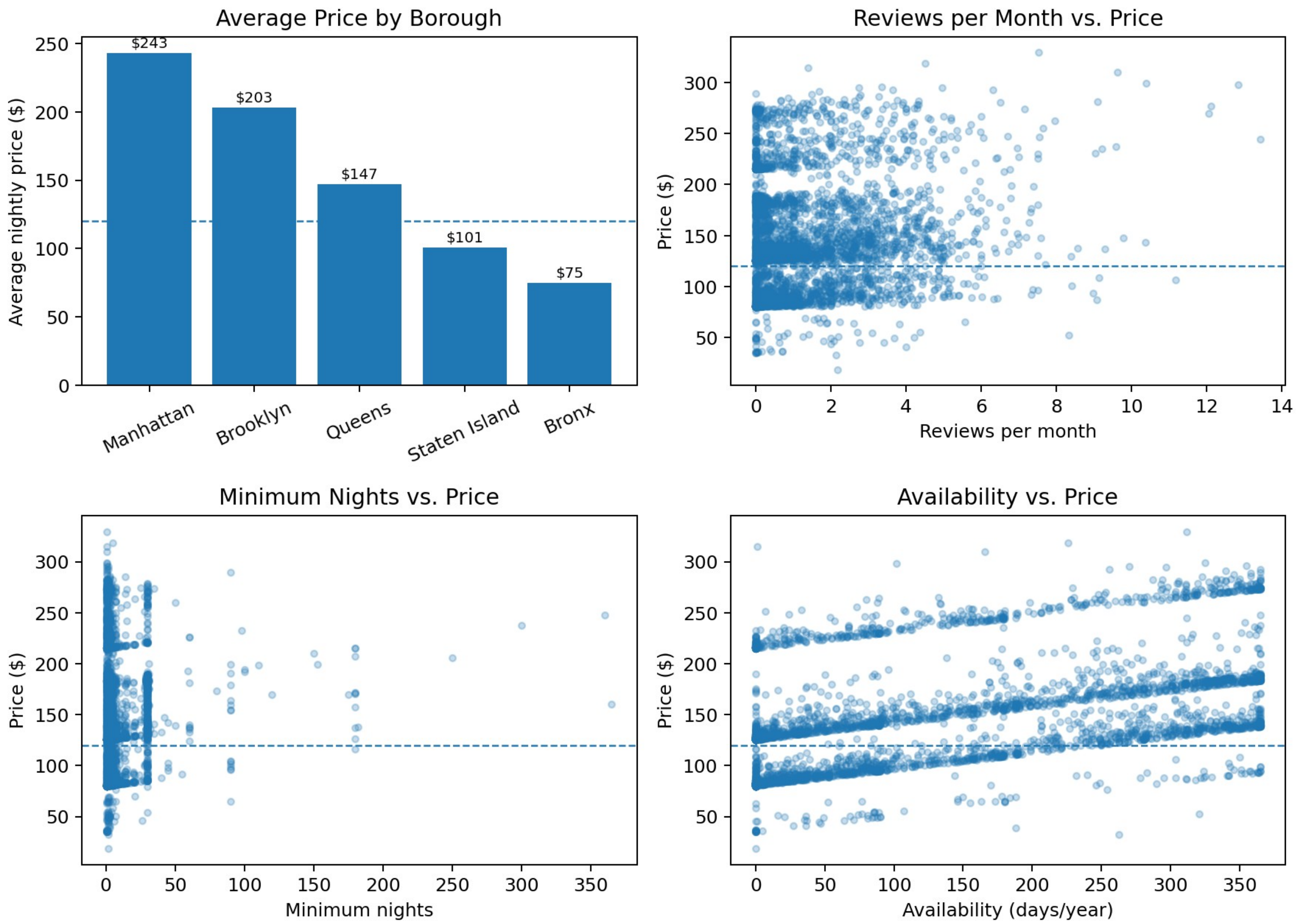
The solution is deployed in Google Colab through a Gradio app with five tabs: Overview, Explore the Data, Model Performance, Coefficient Interpretation, and Price Simulator.

2. Pre-processing and Validation

- Verified all seven required columns were present.
- Converted numeric fields to numeric type and checked for missing values.
- Removed exact duplicate rows.
- Restricted borough codes to 1-5; minimum_nights to at least 1; reviews to nonnegative values; availability to 0-365; and price to positive values.
- Mapped borough codes to readable names and one-hot encoded borough, with Bronx as the reference category.
- Created an above_120 indicator for descriptive KPIs and charts.

Validation result: No rows were removed. The dataset remained at 4,892 records after cleaning.

3. Answers to Joan's Questions



Q1. Which boroughs are associated with higher nightly prices? Manhattan has the highest average price ($242.84), followed by Brooklyn ($202.97), Queens ($147.19), Staten Island ($100.71), and Bronx ($74.78). Borough is the strongest pricing influence.

Q2. Are customer reviews associated with higher rental prices? Yes, but the effect is modest. Each additional review is associated with about $0.15 higher expected price, and each additional review per month with about $0.20, holding other variables constant.

Q3. Do minimum-night requirements influence nightly price? There is a small positive association. Each additional required minimum night is associated with about $0.18 higher expected nightly price.

Q4. Does annual availability affect nightly prices? Yes. Each additional available day is associated with about $0.16 higher expected nightly price, holding the other variables constant. This should be interpreted as an association, not proof of causation.

4. Model Reliability and Overfitting

The data was split 80% training (3,913 rows) and 20% testing (979 rows) using a fixed random seed. The model was fit only on the training data and evaluated on both sets.

Metric	Training	Testing
R^2	0.9812	0.9860
Adjusted R^2	0.9811	0.9859
MAE	$0.75	$0.76
RMSE	$7.18	$5.92

Assessment. The Adjusted R^2 gap is 0.0048. The training and testing values are very close, so there is no meaningful evidence of overfitting in this split. The very high fit should still be treated cautiously because live Airbnb markets include unobserved factors such as exact location, amenities, interior quality, host reputation, and seasonality.

5. Interpretation of Model Coefficients

Variable	Coefficient	Business meaning
Minimum nights	+$0.18	One more required night is associated with a slightly higher expected nightly price.
Total reviews	+$0.15	One additional review is associated with a small increase in expected price.
Reviews per month	+$0.20	One additional monthly review is associated with about $0.20 higher expected price.
Availability	+$0.16	One additional available day is associated with about $0.16 higher expected price.
Manhattan	+$171.59	Expected premium versus a comparable Bronx listing.
Brooklyn	+$124.05	Expected premium versus a comparable Bronx listing.
Queens	+$81.96	Expected premium versus a comparable Bronx listing.
Staten Island	+$36.93	Expected premium versus a comparable Bronx listing.

Conclusion. Borough is the primary screening factor. Manhattan and Brooklyn are the strongest candidates for clearing the $120 threshold, with Queens also performing well. The simulator should support, not replace, due diligence on rent, occupancy, costs, regulations, and landlord authorization.